# Monte Carlo Pricing of a European Call: Control Variates

Quant Finance Portfolio — Phase 2, Block 2, Algorithm 2

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**Abstract**

We apply the control variates technique of the Block 2.0 writeup to the Monte Carlo pricer of the European call under the exact GBM sampler of Block 1.1. Two controls are introduced: the discounted underlying $X_1 = e^{-rT} S_T$ and the discounted asset-or-nothing payoff $X_2 = e^{-rT} S_T \mathbf{1}_{\{S_T > K\}}$. Both have closed-form expectations under GBM. *A naive shape-matching argument suggests $X_2$ should outperform $X_1$ because it matches the call's payoff structurally on both the ITM and OTM regions; this prediction is empirically wrong.* The correct analysis, carried out in detail below, reveals that $X_2$ carries excess variance unrelated to the target and ends up with lower Pearson correlation than $X_1$ (typical values: $\rho_1 \approx 0.92$, $\rho_2 \approx 0.77$). The underlying control delivers VRF $\approx 6.9$ at the ATM point; the AON control delivers only VRF $\approx 2.5$. The lesson generalises beyond this example: structural similarity does not imply linear correlation.

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# 1 Introduction

Block 2.0 derived the control-variate framework: given a target $Y$ with unknown $\mathbb{E}[Y]$ and a control $X$ with known $\mathbb{E}[X]$ correlated with $Y$, the adjusted estimator

$$\tilde{Y}(c) \;=\; Y - c\,(X - \mathbb{E}[X]) \tag{1}$$

is unbiased for $\mathbb{E}[Y]$ for any choice of $c$, with variance

$$\mathrm{Var}(\tilde{Y}(c^*)) \;=\; \mathrm{Var}(Y)(1-\rho^2), \quad c^* \;=\; \frac{\mathrm{Cov}(Y,X)}{\mathrm{Var}(X)}, \quad \rho \;=\; \mathrm{Corr}(Y,X). \tag{2}$$

The variance reduction factor is $\mathrm{VRF} = 1/(1-\rho^2)$, which can be very large when $|\rho|$ is close to 1.

The challenge in applying CV is finding a control. It must satisfy two conditions: (a) its expectation must be known in closed form, (b) it must be highly correlated with the target. The first is restrictive; the second is what distinguishes a good control from a mediocre one.

For the European call under GBM, two controls present themselves naturally: the discounted underlying $X_1 = e^{-rT}S_T$ and the discounted asset-or-nothing payoff $X_2 = e^{-rT}S_T\mathbf{1}_{\{S_T>K\}}$. Both are tractable in closed form. The expectations of these two controls present a clean opportunity to illustrate a subtle but important pedagogical point about how correlation works, which is the main content of this writeup.

# 2 Control 1: the discounted underlying

## 2.1 Definition and tractability

The discounted underlying at maturity is

$$X_1 \;\stackrel{\text{def}}{=}\; e^{-rT}S_T. \tag{3}$$

Under the risk-neutral measure $\mathbb{Q}$, $\{e^{-rt}S_t\}_{t\geq 0}$ is a martingale, so

$$\mathbb{E}^{\mathbb{Q}}[X_1] \;=\; e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T] \;=\; e^{-rT}\cdot S_0 e^{rT} \;=\; S_0. \tag{4}$$

The expectation is therefore known exactly, with no computation needed. This is the simplest possible control variate for any GBM-based pricer.

## 2.2 Naive correlation prediction

The relationship between $Y = e^{-rT}(S_T - K)^+$ and $X_1 = e^{-rT}S_T$ is:

- On $\{S_T > K\}$ (ITM): $Y = X_1 - e^{-rT}K$, linear in $X_1$ with slope 1.
- On $\{S_T \leq K\}$ (OTM): $Y = 0$, while $X_1 > 0$. Here $Y$ is constant and uncorrelated with $X_1$ locally.

A naive prediction based on this dichotomy is that $\rho_1$ should be moderate, perhaps 0.5–0.7, since one of the two regimes (OTM) "wastes" the linear relationship. We will see below that this prediction *underestimates* $\rho_1$ substantially: the actual value at the ATM point is $\rho_1 \approx 0.92$.

# 3 Control 2: the asset-or-nothing payoff

## 3.1 Definition and tractability

The discounted asset-or-nothing call payoff is

$$X_2 \stackrel{\text{def}}{=} e^{-rT} S_T \mathbf{1}_{\{S_T > K\}}. \tag{5}$$

Its expectation has a closed Black-Scholes form. From the standard derivation of the BS call price decomposition,

$$\mathbb{E}^{\mathbb{Q}}[X_2] \;=\; e^{-rT}\mathbb{E}^{\mathbb{Q}}\big[S_T \mathbf{1}_{\{S_T > K\}}\big] \;=\; S_0\,\Phi(d_1), \tag{6}$$

where

$$d_1 \;=\; \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{7}$$

is the standard $d_1$ from Black-Scholes. This identity is the asset-or-nothing building block of the BS formula

$$C^{\text{BS}} \;=\; S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_2 = d_1 - \sigma\sqrt{T}, \tag{8}$$

so $\mathbb{E}[X_2]$ is available for free as a byproduct of the BS pricer code we already implemented in Phase 1.

## 3.2 The naive intuition: "shape matching"

The relationship between $Y$ and $X_2$ is:

- On $\{S_T > K\}$: $Y = X_2 - e^{-rT}K$, linear in $X_2$ with slope 1.
- On $\{S_T \le K\}$: $Y = 0$ *and* $X_2 = 0$ simultaneously, both at the origin.

The naive intuition is that $X_2$ should outperform $X_1$ because it matches the target on *both* regions: zero where the target is zero, and linearly related where the target is linear. The OTM "wasted region" that hurt $X_1$ does not exist for $X_2$; instead both target and control vanish together. One might therefore predict $\rho_2$ near 1, perhaps above 0.95.

*This prediction is wrong*, and the rest of this writeup is devoted to understanding why. Structural similarity — vanishing together, linear together — is not the same as global linear correlation, and the difference is exactly what Pearson's $\rho$ measures.

# 4 The empirical surprise and the correct analysis

We computed $\rho_1$ and $\rho_2$ analytically (and confirmed empirically with $10^6$ paths) for the canonical ATM parameters ($S_0 = K = 100, r = 0.05, \sigma = 0.20, T = 1$). The results:

$$\rho_1 \;=\; 0.9245, \qquad \rho_2 \;=\; 0.7714, \tag{9}$$

$$\text{VRF}_1 \;=\; \frac{1}{1-\rho_1^2} \;=\; 6.88, \qquad \text{VRF}_2 \;=\; \frac{1}{1-\rho_2^2} \;=\; 2.47. \tag{10}$$

The underlying control beats the asset-or-nothing control by almost a factor of 3 in variance reduction. The naive prediction was inverted: $X_1$ is the better control, not $X_2$.

### 4.1 The closed-form analysis

The key algebraic observation is that $Y$ and $X_2$ differ by a deterministic function of the path:

$$Y \;=\; X_2 - e^{-rT}K \cdot A, \qquad A \;\stackrel{\text{def}}{=}\; \mathbf{1}_{\{S_T > K\}}. \tag{11}$$

On $\{S_T > K\}$ both sides equal $e^{-rT}(S_T - K)$; on $\{S_T \le K\}$ both sides equal zero. So $Y$ is exactly $X_2$ shifted down by a Bernoulli random variable scaled by the discounted strike.

From this identity:

$$\mathrm{Var}(Y) \;=\; \mathrm{Var}(X_2) - 2e^{-rT}K\ \mathrm{Cov}(X_2, A) + (e^{-rT}K)^2\,\mathrm{Var}(A), \tag{12}$$

$$\mathrm{Cov}(Y, X_2) \;=\; \mathrm{Var}(X_2) - e^{-rT}K\ \mathrm{Cov}(X_2, A). \tag{13}$$

The closed forms for the second moments under GBM (lognormal MGF applied to the truncated regions) are:

$$\mathbb{E}[X_1] = S_0, \qquad \mathrm{Var}(X_1) = S_0^2\big(e^{\sigma^2 T} - 1\big), \tag{14}$$

$$\mathbb{E}[X_2] = S_0\Phi(d_1), \qquad \mathbb{E}[X_2^2] = S_0^2\, e^{\sigma^2 T}\,\Phi(d_1 + \sigma\sqrt{T}), \tag{15}$$

and the cross terms $\mathrm{Cov}(X_1, A)$, $\mathrm{Cov}(X_2, A)$, $\mathrm{Cov}(X_1, X_2)$ likewise have closed forms in terms of $\Phi(d_1)$, $\Phi(d_2)$, and $\Phi(d_1 + \sigma\sqrt{T})$. We omit the routine algebra and quote the numerical values for our parameters:

$$\mathrm{Var}(Y) = 216.66, \quad \mathrm{Var}(X_1) = 408.11, \quad \mathrm{Var}(X_2) = 3322.15. \tag{16}$$

The most informative comparison is between $\mathrm{Var}(X_1)$ and $\mathrm{Var}(X_2)$. The variance of the asset-or-nothing control is *eight times* that of the underlying control, and *fifteen times* that of the target itself.

### 4.2 Why excess variance hurts correlation

Pearson's correlation is

$$\rho \;=\; \frac{\mathrm{Cov}(Y, X)}{\sqrt{\mathrm{Var}(Y)\cdot\mathrm{Var}(X)}}. \tag{17}$$

The covariance and the target variance are fixed by the problem; the control's variance enters in the denominator. *A control with excess variance — variance not explained by linear association with $Y$ — gets penalised in the correlation coefficient.*

In our case, the algebraic source of excess variance for $X_2$ is the multiplicative structure of the indicator $\mathbf{1}_{\{S_T>K\}}$: when the call is ITM, the value of $X_2$ depends on $S_T$, which ranges widely (the typical ITM path can have $S_T$ anywhere from $K$ to several times $K$); the variance of $X_2$ on the ITM region is therefore large. The target $Y = X_2 - e^{-rT}K$ on the same region has the *same* variance as $X_2$ on this region, but the *global* target variance is much smaller because $Y = 0$ on OTM (a region of probability $\approx 0.44$). The denominator of $\rho_2$ therefore picks up the full $\mathrm{Var}(X_2)$ while the numerator $\mathrm{Cov}(Y, X_2)$ picks up only the portion correlated with $Y$.

For $X_1$, the analogous algebra plays out differently. $X_1$ has *positive but limited* variance on both ITM and OTM regions: the variance on OTM is constrained because $S_T \in (0, K]$ there. This bounded behaviour makes $\mathrm{Var}(X_1)$ small, only $1.9 \times \mathrm{Var}(Y)$, so the denominator does not penalise the correlation heavily.

### 4.3 The optimal coefficient confirms the picture

Another way to see the asymmetry is via the optimal coefficient $c^*$:

$$c_1^* = \frac{\mathrm{Cov}(Y, X_1)}{\mathrm{Var}(X_1)} = \frac{274.91}{408.11} = 0.674, \tag{18}$$

$$c_2^* = \frac{\mathrm{Cov}(Y, X_2)}{\mathrm{Var}(X_2)} = \frac{654.44}{3322.15} = 0.197. \tag{19}$$

For the underlying, $c_1^* \approx 0.67$, indicating that subtracting $0.67 \times (X_1 - S_0)$ from $Y$ removes about two-thirds of the "directional" variance. For the AON, $c_2^* \approx 0.20$, much smaller: the algorithm "does not trust" the AON nearly as much as it does the underlying, because the AON has so much variance unrelated to $Y$.

## 5 The general lesson

The mistake corrected in Section 4 is a subtle one. The naive intuition — that $X_2$ matches $Y$ on both regions — is correct: $Y$ and $X_2$ are zero together and linear together. But this kind of *piecewise* matching is not what Pearson's correlation measures; Pearson's correlation measures *global linear fit*. Two variables can match each other piecewise and still have moderate correlation if the piecewise pieces have very different variances.

A useful diagnostic that would have revealed the problem before running any code: compute the ratio $\mathrm{Var}(Y)/\mathrm{Var}(X)$ for each candidate control. A good control has this ratio close to 1: the control varies on the same scale as the target. A control with $\mathrm{Var}(Y)/\mathrm{Var}(X) \ll 1$ has excess variance and will likely give modest $\rho$.

| Control | $\mathrm{Var}(Y)/\mathrm{Var}(X)$ | Empirical $\rho$ | VRF |
|---|---|---|---|
| $X_1$ (underlying) | 0.53 | 0.92 | 6.88 |
| $X_2$ (AON) | 0.07 | 0.77 | 2.47 |

The variance-ratio diagnostic correctly predicts the ranking of the two controls. The naive shape-matching argument does not.

## 6 Implementation: empirical $\hat{c}$

We use the empirical estimator of $c^*$ from the same paths that produce the estimator. Given samples $\{(Y_i, X_i)\}_{i=1}^M$:

$$\hat{c} = \frac{\sum_{i=1}^M (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_{i=1}^M (X_i - \bar{X})^2} \quad \text{(OLS slope of } Y \text{ on } X\text{)}, \tag{20}$$

$$\hat{C}_M^{\mathrm{CV}} = \frac{1}{M} \sum_{i=1}^M \big(Y_i - \hat{c}\,(X_i - \mathbb{E}[X])\big). \tag{21}$$

The bias of $\hat{C}_M^{\mathrm{CV}}$ is $O(1/M)$, while the standard error is $O(1/\sqrt{M})$, so the bias is asymptotically negligible for any $M$ at which the standard error is acceptable. See Block 2.0 writeup, Proposition 4.4.

### 6.1 Connection with OLS regression

The slope estimator $\hat{c}$ is exactly the slope of the OLS regression of $Y$ on $X$ with an intercept. The CV-adjusted samples $\tilde{Y}_i = Y_i - \hat{c}(X_i - \mathbb{E}[X])$ are the regression residuals shifted by $\hat{\alpha} + \hat{c}\mathbb{E}[X]$ to recover the original mean. The variance of $\tilde{Y}_i$ is the variance of the regression residuals, which is $\mathrm{Var}(Y)(1 - R^2)$ with $R^2 = \rho^2$ for a single regressor. The CV variance reduction is therefore exactly the OLS coefficient of determination.

# 7 Algorithmic specification

**Algorithm 1** Control-variate Monte Carlo for the European call (exact GBM)

**Require:** Spot $S_0$, strike $K$, maturity $T$, rate $r$, volatility $\sigma$, sample size $M$, control identifier $c \in \{1, 2\}$, confidence level $1 - \beta$, random generator `rng`.

**Ensure:** Estimate $\hat{C}^{\mathrm{CV}}$, half-width $h_M^{\mathrm{CV}}$, sample variance $S_M^{2,\mathrm{CV}}$.

$D \leftarrow e^{-rT}$
**for** $i = 1, \ldots, M$ **do**
  Draw $Z_i \sim \mathcal{N}(0, 1)$ using `rng`
  $S_T^{(i)} \leftarrow S_0 \exp((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T} Z_i)$
  $Y_i \leftarrow D \cdot \max(S_T^{(i)} - K,\, 0)$
  **if** $c = 1$ ▷ control: discounted underlying **then**
    $X_i \leftarrow D \cdot S_T^{(i)}, \quad \mathbb{E}X \leftarrow S_0$
  **else if** $c = 2$ ▷ control: asset-or-nothing **then**
    $X_i \leftarrow D \cdot S_T^{(i)} \cdot \mathbf{1}_{\{S_T^{(i)} > K\}}$
    $\mathbb{E}X \leftarrow S_0\, \Phi(d_1)$ ▷ closed BS form
  **end if**
**end for**
$\bar{Y} \leftarrow M^{-1} \sum_i Y_i, \quad \bar{X} \leftarrow M^{-1} \sum_i X_i$
$\hat{c} \leftarrow \frac{\sum_i (Y_i - \bar{Y})(X_i - \bar{X})}{\sum_i (X_i - \bar{X})^2}$ ▷ OLS slope
**for** $i = 1, \ldots, M$ **do**
  $\tilde{Y}_i \leftarrow Y_i - \hat{c}\,(X_i - \mathbb{E}X)$
**end for**
Reduce: $\hat{C}^{\mathrm{CV}} \leftarrow M^{-1} \sum_i \tilde{Y}_i, \quad S_M^{2,\mathrm{CV}} \leftarrow (M-1)^{-1} \sum_i (\tilde{Y}_i - \hat{C}^{\mathrm{CV}})^2$
$h_M^{\mathrm{CV}} \leftarrow z_{\beta/2} \sqrt{S_M^{2,\mathrm{CV}}/M}$
**return** $(\hat{C}^{\mathrm{CV}},\, h_M^{\mathrm{CV}},\, S_M^{2,\mathrm{CV}})$

# 8 Empirical validation strategy

The validation script for the CV pricers comprises four tests, run separately for each control.

**Test 1: empirical correlation and VRF.** For each control, generate $10^6$ paths and measure $\hat{\rho} = \widehat{\mathrm{Corr}}(Y, X)$ and $\widehat{\mathrm{VRF}} = 1/(1 - \hat{\rho}^2)$. The *corrected* predictions, based on the analysis of Section 4:

- Control 1 (underlying): $\hat{\rho}_1 \approx 0.92$, $\mathrm{VRF}_1 \approx 6.9$.
- Control 2 (AON): $\hat{\rho}_2 \approx 0.77$, $\mathrm{VRF}_2 \approx 2.5$.

The control 1 (underlying) is the better choice for European call pricing under GBM, contrary to the naive shape-matching prediction.

**Test 2: BS coherence.** Each pricer at $M = 50\,000$ paths should agree with $C^{\mathrm{BS}}$ within a few half-widths.

**Test 3: half-width comparison vs IID at fixed budget.** For each control, run the CV pricer with $M = 100\,000$ paths and the IID pricer with the same $M = 100\,000$ paths from the same seed. The squared half-width ratios IID/CV should match the VRFs from Test 1.

**Test 4: input validation.** Mechanical: both pricers reject bad inputs and accept negative $r$.

## 9 Bibliographic notes

The textbook reference is Glasserman [2, Section 4.1]. The asset-or-nothing decomposition of the BS formula is in Hull [3, Chapter 26]. The OLS interpretation of the optimal control coefficient is well-treated in Asmussen & Glynn [1, Chapter V]. The pedagogical point about shape matching versus correlation, while well-known to practitioners, does not appear to be highlighted in any of the standard textbook sources we consulted.

## References


[1] S. Asmussen and P. W. Glynn, *Stochastic Simulation: Algorithms and Analysis*, Springer, 2007.

[2] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[3] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th edition, Pearson, 2018.